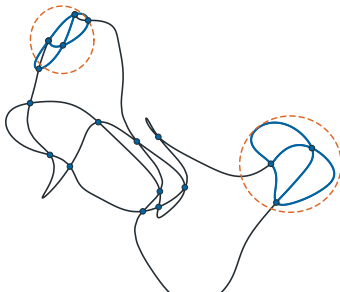


A Slots selected from the shadow

On $\mathcal{G}_n$, the shadow determines $s_n = \lfloor \delta n \rfloor$ pairwise separated opened-trefoil slots.



slots chosen before internal signs are exposed

schematic drawing: $n = 20$, $s_n = 2$, and $s_n/n = 1/10$

B Connected-sum factors and anti-concentration

The tangle circles remain disjoint under every internal redecoration.
Conditional on $\mathcal{F}_n$, internal signs independently determine the local unknot or trefoil factors.

each selected slot

$2/8$ close to $T(2, \pm 3)$
(trefoil mirror pair)

$6/8$ close to $T(2, \pm 1)$
(unknot)

$$Y_{n,j} = \mathbf{1}_{\{L_{n,j} \text{ trefoil}\}}$$

$$Y_{n,j} \mid \mathcal{F}_n \sim \text{Bernoulli}(1/4)$$

on $\mathcal{G}_n$; conditionally independent across slots

determinant 3-adic valuation

$$K(D_n) \cong K_n^{\text{ext}} \# L_{n,1} \# \cdots \# L_{n,s_n}$$

K_n^{ext} is fixed after the shadow and exterior signs are exposed; each $L_{n,j}$ is an unknot or trefoil.

$$\mathcal{L}(B_n \mid \mathcal{F}_n) = \text{Bin}(s_n, 1/4) \quad \text{on } \mathcal{G}_n$$

$$\det K(D_n) = \det(K_n^{\text{ext}}) 3^{B_n}$$

$$v_3(\det K(D_n)) = v_3(\det K_n^{\text{ext}}) + B_n$$

$$\det K(D_n) = a \implies B_n = v_3(a) - v_3(\det K_n^{\text{ext}})$$

each determinant target permits at most one value of B_n

moving target

$$\mathbb{P}\{\det = a \mid \mathcal{F}_n\}$$

$$\leq \max_b \mathbb{P}\{B_n = b \mid \mathcal{F}_n\}$$

$$= O(s_n^{-1/2}) = O(n^{-1/2}) \quad \text{on } \mathcal{G}_n$$

fixed target

$$\det K(D_n) = a_* \implies B_n \leq v_3(a_*)$$

$$\mathbb{P}\{\det = a_*\} = O(e^{-c_* n})$$